

DBS Assignment 5

Normalization of Toy Store data

Hand-in info

- Hand in one .pdf file with your answers to the tasks below.

Introduction:

Some more toy stores have joined the PlayDate group, and in an attempt to not lose any of the sales data, it has been decided to paste it all into a big table. This is some of the data in the “toy_store” table that was the result of that:

storeName	storeAddress	receiptNo	saleTime	saleItems (sku, itemName, qty, promoCode, promoName, warrantyCode, warrantyMonths)	clerkId	clerkName
PlayWorld	Other Street 3	135	2025-08-26 10:12	[[101, LEGO, 1, PR10, Summer, WR10, 10], {105, yoyo, 1, PR20, MoreSummer, WR10, 10}]	C11	Peter
PlayWorld	Other Street 3	266	2025-08-26 10:27	[[82, puzzle, 1, PR22, BackToSchool, WR1, 1]]	C11	Peter
ToyGalaxy	MyWay Rd 5	266	2025-08-26 11:05	[[85, Drone, 2, PR22, BackToSchool, WR5, 5]]	C24	Axel
ToyGalaxy	MyWay Rd 5	222	2025-08-26 11:40	[[85, Drone, 1, PR22, BackToSchool, WR5, 5]]	C02	Christian
FunTown	Any Ave 6	101	2025-08-26 12:10	[[82, puzzle, 1, PR22, BackToSchool, WR1, 1], {105, yoyo, 1, PR20, MoreSummer, WR10, 10}]	C07	Rosa
FunTown	Any Ave 6	200	2025-08-26 13:15	[[105, yoyo, 1, PR20, MoreSummer, WR10, 10]]	C07	Rosa

Task A: anomalies

Before you get to fix this mess of a table, then list some of the anomalies here, that could happen when working with the table in its current form.

Use concrete examples and mention which type of anomaly it is (insert/update/delete).

Task B: 1NF

Expand the table with any extra columns and/or rows necessary to get it to be on 1NF.

Then show the result here.

Task C: functional dependencies

List all the functional dependencies that you can identify in your new table, and include the type (full/partial/transitive).

Use the concrete column names and show the dependencies using the arrow notation.

Task D: keys

List all possible candidate keys in your new table, and mark which one you have chosen as your primary key.

Explain your choice of primary key.

Task E: 2NF

Get your table to 2NF by getting rid of any partial dependencies and show your resulting table(s) here.

Explain any interesting choices and/or decisions you have made.

Task F: 3NF

Get your table(s) to 3NF by getting rid of any transitive dependencies and show your resulting table(s) here.

Explain any interesting choices and/or decisions you have made.